

Water, Air, & Soil Pollution

Long-term genomic surveillance unveils key drivers of Salmonella contamination and antimicrobial resistance dynamics in surface waters from Central Mexico

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Online Resource 2. Consolidated output of GLMMs validation of Salmonella prevalence and MDR phenotypes

SECTION 1. RAW GLMM CONSOLE SUMMARY: GLOBAL SALMONELLA PREVALENCE

Generalized linear mixed model fit by maximum likelihood (Laplace Approximation) ['glmerMod']
Family: binomial (logit)
Formula: Result ~ Year_Fact + State + Source + Season + pH + Temperature + Turbidity + (1 | Site)
Data: base_muestras_anual

AIC	BIC	logLik	-2*log(L)	df.resid
1091.4	1182.6	-526.7	1053.4	880

Scaled residuals:

Min	1Q	Median	3Q	Max
-3.6254	-0.8347	0.3601	0.7448	4.0329

Random effects:

Groups Name	Variance	Std.Dev.
Site (Intercept)	0	0

Number of obs: 899, groups: Site, 49

Fixed effects:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	5.843648	1.232289	4.742	2.12e-06	***
Year_Fact2020	-0.896919	0.331676	-2.704	0.00685	**
Year_Fact2021	-0.568689	0.260695	-2.181	0.02915	*
Year_Fact2022	-1.349833	0.225710	-5.980	2.23e-09	***

Year_Fact2023	-2.498690	0.279426	-8.942	< 2e-16	***
StateMexico City	0.209400	0.439730	0.476	0.63393	
StateMorelos	0.488631	0.332146	1.471	0.14125	
StateState of Mexico	0.550876	0.383632	1.436	0.15102	
StateTlaxcala	1.099102	0.394976	2.783	0.00539	**
SourceDam	-0.045899	0.383617	-0.120	0.90476	
SourcePond	-0.414559	0.298051	-1.391	0.16426	
SourceRiver	-0.041161	0.242957	-0.169	0.86547	
SeasonSpring	1.928028	0.256634	7.513	5.79e-14	***
SeasonSummer	0.249867	0.210144	1.189	0.23443	
SeasonWinter	1.374387	0.251216	5.471	4.48e-08	***
pH	-0.768214	0.143448	-5.355	8.54e-08	***
Temperature	0.007104	0.020838	0.341	0.73316	
Turbidity	0.001204	0.001372	0.878	0.37992	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
 optimizer (Nelder_Mead) convergence code: 0 (OK)
 boundary (singular) fit: see help('isSingular')

SECTION 2. PARALLEL AUDIT MATRIX: STANDARD GLM PREVALENCE DRIVERS
 Executed systematically to validate fixed effects under zero random
 variance thresholds

Call:

```
glm(formula = Result ~ Year_Fact + State + Source + Season +
    pH + Temperature + Turbidity, family = binomial(link = "logit"),
    data = base_muestras_anual)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	5.843649	1.232406	4.742	2.12e-06	***
Year_Fact2020	-0.896919	0.331678	-2.704	0.00685	**
Year_Fact2021	-0.568689	0.260695	-2.181	0.02915	*
Year_Fact2022	-1.349833	0.225712	-5.980	2.23e-09	***
Year_Fact2023	-2.498691	0.279427	-8.942	< 2e-16	***
StateMexico City	0.209400	0.439737	0.476	0.63394	
StateMorelos	0.488631	0.332149	1.471	0.14126	
StateState of Mexico	0.550876	0.383639	1.436	0.15102	
StateTlaxcala	1.099102	0.394981	2.783	0.00539	**
SourceDam	-0.045899	0.383620	-0.120	0.90476	
SourcePond	-0.414559	0.298053	-1.391	0.16426	
SourceRiver	-0.041161	0.242958	-0.169	0.86547	
SeasonSpring	1.928028	0.256635	7.513	5.79e-14	***
SeasonSummer	0.249867	0.210145	1.189	0.23443	
SeasonWinter	1.374387	0.251217	5.471	4.48e-08	***
pH	-0.768215	0.143458	-5.355	8.56e-08	***
Temperature	0.007104	0.020838	0.341	0.73316	

Turbidity 0.001204 0.001372 0.878 0.37993

 Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 1241.9 on 898 degrees of freedom
 Residual deviance: 1053.4 on 881 degrees of freedom
 AIC: 1089.4

Number of Fisher Scoring iterations: 4

 SECTION 3. RAW GLMM CONSOLE SUMMARY: GLOBAL MDR PHENOTYPES
 Generated automatically via open-science verification pipelines

Generalized linear mixed model fit by maximum likelihood (Laplace
 Approximation) ['glmerMod']
 Family: binomial (logit)
 Formula: MDRf ~ Year_Fact + State + Source + Season + pH + Temperature
 + Turbidity + (1 | Site)
 Data: base_mdrrf_consolidada

AIC	BIC	logLik	-2*log(L)	df.resid
607.3	687.7	-284.6	569.3	489

Scaled residuals:

Min	1Q	Median	3Q	Max
-1.5579	-0.6741	-0.5070	1.0346	2.6037

Random effects:

Groups Name	Variance	Std.Dev.
Site (Intercept)	0	0

Number of obs: 508, groups: Site, 47

Fixed effects:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.131256	1.734400	0.076	0.93968
Year_Fact2020	-1.112600	0.562556	-1.978	0.04796 *
Year_Fact2021	0.087808	0.417000	0.211	0.83322
Year_Fact2022	-0.306451	0.324130	-0.945	0.34443
Year_Fact2023	-0.449099	0.420151	-1.069	0.28512
StateMexico City	-1.206395	0.814342	-1.481	0.13849
StateMorelos	-1.925284	0.719431	-2.676	0.00745 **
StateState of Mexico	-1.323501	0.722040	-1.833	0.06680 .

StateTlaxcala	-1.596102	0.738039	-2.163	0.03057 *
SourceDam	-0.839463	0.766708	-1.095	0.27356
SourcePond	0.099267	0.497638	0.199	0.84189
SourceRiver	0.595892	0.315000	1.892	0.05853 .
SeasonSpring	0.441137	0.355255	1.242	0.21433
SeasonSummer	-0.087465	0.310644	-0.282	0.77828
SeasonWinter	0.624384	0.423451	1.475	0.14034
pH	-0.002155	0.201310	-0.011	0.99146
Temperature	-0.002259	0.029697	-0.076	0.93937
Turbidity	0.002777	0.001600	1.736	0.08252 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
optimizer (Nelder_Mead) convergence code: 0 (OK)
boundary (singular) fit: see help('isSingular')

SECTION 4. PARALLEL AUDIT MATRIX: STANDARD GLM MDRf DRIVERS
Executed systematically to validate isolate fixed effects under zero
random variance thresholds

Call:
glm(formula = MDRf ~ Year_Fact + State + Source + Season + pH +
Temperature + Turbidity, family = binomial(link = "logit"),
data = base_isolates_audit)

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.131256	1.734453	0.076	0.93968
Year_Fact2020	-1.112600	0.562554	-1.978	0.04796 *
Year_Fact2021	0.087808	0.417001	0.211	0.83322
Year_Fact2022	-0.306451	0.324132	-0.945	0.34443
Year_Fact2023	-0.449099	0.420150	-1.069	0.28511
StateMexico City	-1.206395	0.814302	-1.482	0.13847
StateMorelos	-1.925284	0.719381	-2.676	0.00744 **
StateState of Mexico	-1.323501	0.721998	-1.833	0.06679 .
StateTlaxcala	-1.596102	0.737996	-2.163	0.03056 *
SourceDam	-0.839463	0.766660	-1.095	0.27353
SourcePond	0.099267	0.497636	0.199	0.84189
SourceRiver	0.595892	0.315000	1.892	0.05853 .
SeasonSpring	0.441137	0.355255	1.242	0.21433
SeasonSummer	-0.087465	0.310645	-0.282	0.77828
SeasonWinter	0.624384	0.423450	1.475	0.14034
pH	-0.002155	0.201317	-0.011	0.99146
Temperature	-0.002259	0.029697	-0.076	0.93937
Turbidity	0.002777	0.001600	1.736	0.08252 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 607.58 on 507 degrees of freedom
Residual deviance: 569.28 on 490 degrees of freedom
AIC: 605.28

Number of Fisher Scoring iterations: 4